Assignment 5

https://www.notion.so/Assignment-5-20cee40c6a088002b336f00bd583a51c?source=copy_link

1. Write method headers (not the bodies) for the following methods:
    a. Return a sales commision, given the sales amount and the commision rate.

    ```
    public static double salesCommision(double salesAmount, double commi
    ```

    b. Display the calendar for a month, given the month and year.

    ```
    public static void displayCalendar(int month, int year)
    ```

    c. Return a square root of a number.

    ```
    public static double root(int number)
    ```

    d. Test whether a number is even, and returning true if it is.

    ```
    public static boolean isEven(int number)
    ```

    e. Display a message a specified number of times.

    ```
    public static void message(int number)
    ```

    f. Return the monthly payment, given the loan amount, number of years, and annual interest rate.

    ```
    public static double payment(double loanAmount, int years,
        double interestRate)
    ```

    g. Return the corresponding uppercase letter, given a lowercase letter.

```
public static char toUpperCase(char letter)
```

2. Identify and correct the errors in the following program:

```
1. public static void method1 ...
2. method2(int(3.4))
```

3. What is pass-by-value? Show the result of the following programs.

   i. 0

   ii.
   ```
   2
   2 4
   2 4 8
   2 4 8 16
   2 4 8 16 32
   2 4 8 16 32 64
   ```

   iii.
   ```
   Before the call,variable times is 3
   n = 3
   Welcome to Java!
   n = 2
   Welcome to Java!
   n = 1
   Welcome to Java!
   After the call,variable times is 3
   ```

   iv.
   ```
   1
   2 1
   2 1
   4 2 1
   i is 5
   ```

4. What is wrong in the following program?

```
method is already defined
```

5. Given two method definitions, tell which of the two methods is invoked for:
   i. second
   ii. second
   iii. first
6. Write a method that displays an n by n matrix using the following header:

```
class Main {
   public static void main(String[] args) {
      printMatrix(3);
   }
   public static void printMatrix(int n) {
     for (int i = 0; i < n; i++) {
       for (int j = 0; j < n; j++) {
         // Generate either 0 or 1 randomly
         int randomBit = (int)(Math.random() * 2); // 0 or 1
         System.out.print(randomBit + " ");
       }
       System.out.println(); // Move to next row
     }
   }
}
```

7. Write a method name grade(), that accepts one parameter named score (typed double). The method returns a grade based on the score.

```
class Main {
   public static void main(String[] args) {
      double score = 72.5;
      char result = grade(score);
      System.out.println("Score: " + score + " → Grade: " + result);
   }
```

```
    public static char grade(double score) {
        if (score >= 80.0) {
            return 'A';
        } else if (score >= 65.0) {
            return 'B';
        } else if (score >= 50.0) {
            return 'C';
        } else if (score >= 40.0) {
            return 'D';
        } else {
            return 'F';
        }
}
}
```

8. Modify the program code to fulfill the following requirements.

```
class Main {
  public static void main(String[] args) {
    double weightInKg = 53.5;
    double heightInMeter = 1.68;

    double bmi = calcBmi(weightInKg, heightInMeter);

    System.out.println("The BMI is: " + bmi);
  }

  public static double calcBmi(double weightInKg, double heightInMeter) {
    double bmi = weightInKg / (heightInMeter * heightInMeter);
    return bmi;
  }
}
```

9. Given below is a method that receives 3 parameters of integer type. This method returns the sum of all integers to the caller. Using the method given, write codes in main method to declare, call the method and display the calculated sum of the result in the method.

```
class Main {

  public static void main(String[] args) {
    int a = 1;
    int b = 2;
    int c = 3;

    int total = calcTotal(a, b, c);

    System.out.println("The total sum is: " + total);
  }

  public static int calcTotal(int x, int y, int z) {
    return x + y + z;
  }
}
```